Mohammed Sinan

Bangalore, India | +91 9538887428 | mohammedsinan9705@gmail.com | github.com/Mdsinan09 | linkedin.com/in/mdsinan09

Profile

Backend and cloud-focused engineer who builds production-style distributed systems and cloud infrastructure — API platforms, real-time systems, and multi-region deployment pipelines. Hands-on experience across Node.js, PostgreSQL, Redis, Docker, Kubernetes, and AWS (EC2, S3, RDS), with full-stack exposure through React.

Education

KS Institute of Technology Bangalore, India
Bachelor of Engineering in Computer Science & Design *2023 – 2027 (Expected)*

Technical Skills

Languages: JavaScript, TypeScript, Python, Java, C

Frontend / Backend: React.js, Tailwind CSS, Node.js, Express.js, REST APIs

Databases: PostgreSQL, MongoDB

Cloud & DevOps: Kubernetes (K3s), Docker, AWS (EC2, S3, RDS), GitHub Actions, Cloudflare Workers

Other: WebSocket, OpenAI API, System Design, Git

Projects

Cloud-Native Multi-Region Deployment Platform | *Node.js, TypeScript, Kubernetes (K3s), Docker, PostgreSQL* GitHub

- Built a GitOps-style deployment platform that builds and deploys apps automatically on every GitHub push, using HMAC-verified webhooks, commit-based idempotency, and Redis-backed per-app deployment locking via a BullMQ job queue
- Automated the build pipeline (clone → Docker build → registry push) and deployed generated Kubernetes manifests in parallel across two independent K3s clusters representing separate regions
- Implemented health-check-gated rollouts with deterministic rollback: failed health checks automatically trigger re-application of the last successful manifest stored in PostgreSQL across all regions
- Built a Cloudflare Worker for geo-aware traffic routing and regional failover, plus a React dashboard showing live deployment state, pod status, streaming logs, and Prometheus metrics

AI-Powered API Testing & Mocking Platform | *React, Node.js, OpenAI API, PostgreSQL, Docker* GitHub

- Built a developer tool that auto-generates realistic API mocks from OpenAPI schemas using OpenAI structured outputs, removing the need for manual mock setup
- Designed a dynamic mock server that spins up live Express endpoints at runtime from parsed schema definitions
- Added configurable load testing with virtual users and a real-time dashboard for latency percentiles and throughput

AI Assistive Tutor for Visually Impaired Students | *Python, FastAPI, PyTorch, OpenCV, Tesseract OCR, SQLite* GitHub

- Developed a voice-first assistive learning platform integrating real-time object detection, OCR-based textbook reading, and contextual AI tutoring for visually impaired students
- Implemented a FieldNet V3 object-detection pipeline with spatial reasoning to identify study objects and provide directional audio guidance; built an OpenCV and Tesseract OCR pipeline to extract and read printed study material
- Connected scanned text to a Gemini-powered academic tutor with an extractive fallback, and integrated voice-command intent routing, speech feedback, a 2D digital twin, and session analytics

Certifications

JP Morgan Quantitative Research Job Simulation (Forage, 2026) — commodity pricing models and credit risk analysis

Deloitte Technology Job Simulation (Forage, 2026) — data normalization and a client-facing development proposal

Deloitte Data Analytics Job Simulation (Forage, 2026) — forensic technology and data analysis